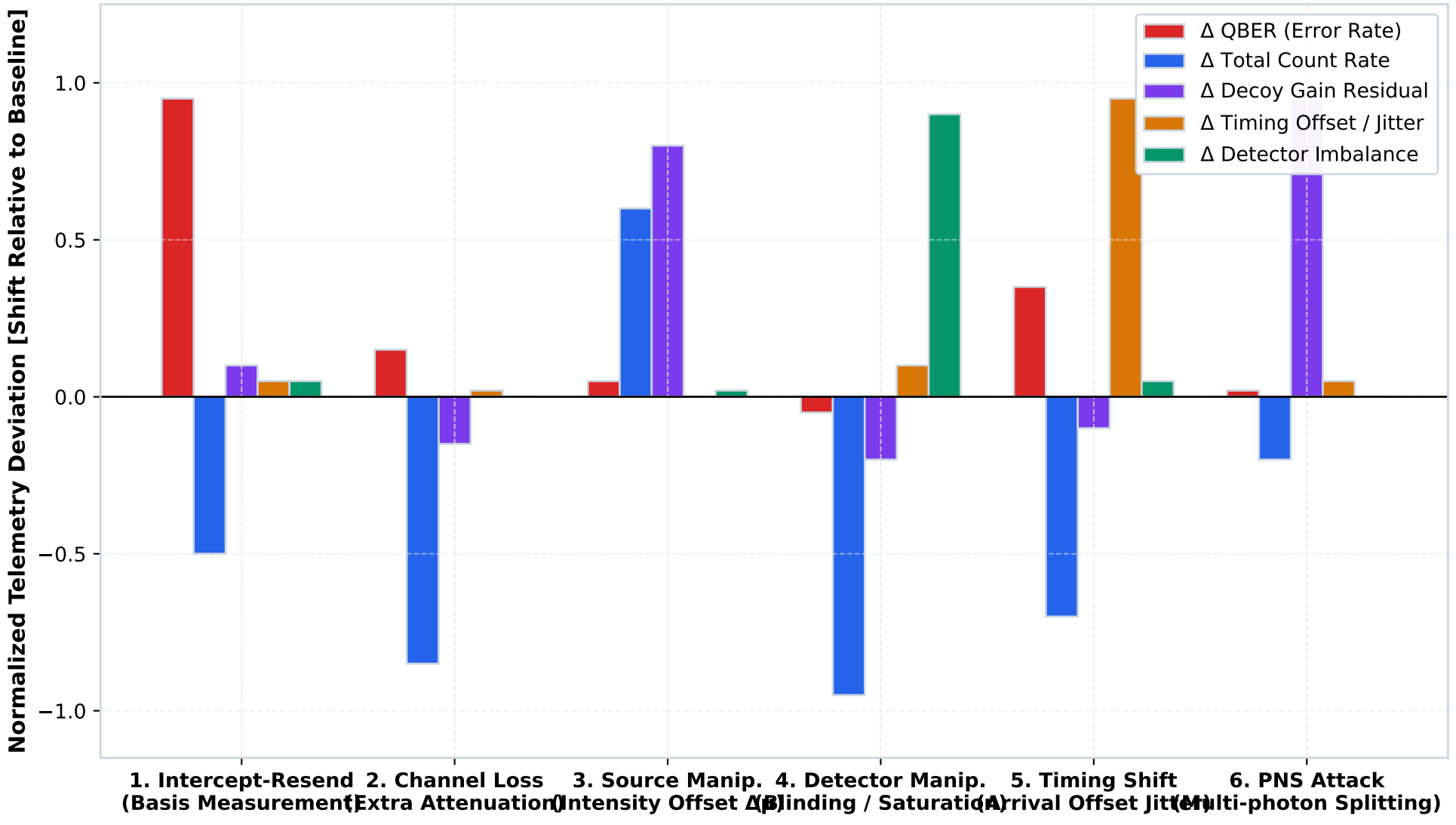
Multi-Attack Physical Footprints: Distinct Observational Signatures Across Telemetry Dimensions
Normalized Telemetry Deviation [Shift Relative to Baseline]
1.0
0.5
0.0
−0.5
−1.0
Δ QBER (Error Rate)
Δ Total Count Rate
Δ Decoy Gain Residual
Δ Timing Offset / Jitter
Δ Detector Imbalance
1. Intercept-Resend (Basis Measurement)
2. Channel Loss (Extra Attenuation)
3. Source Manip. (Intensity Offset Δ(B)
4. Detector Manip. (Blinding / Saturation)
5. Timing Shift (Arrival Offset Jitter)
6. PNS Attack (Multi-photon Splitting)